机电系统建模、辨识与控制
Modeling, Identification and Control of Mechatronic Systems

MIC_L3 Principle of Feedback Control

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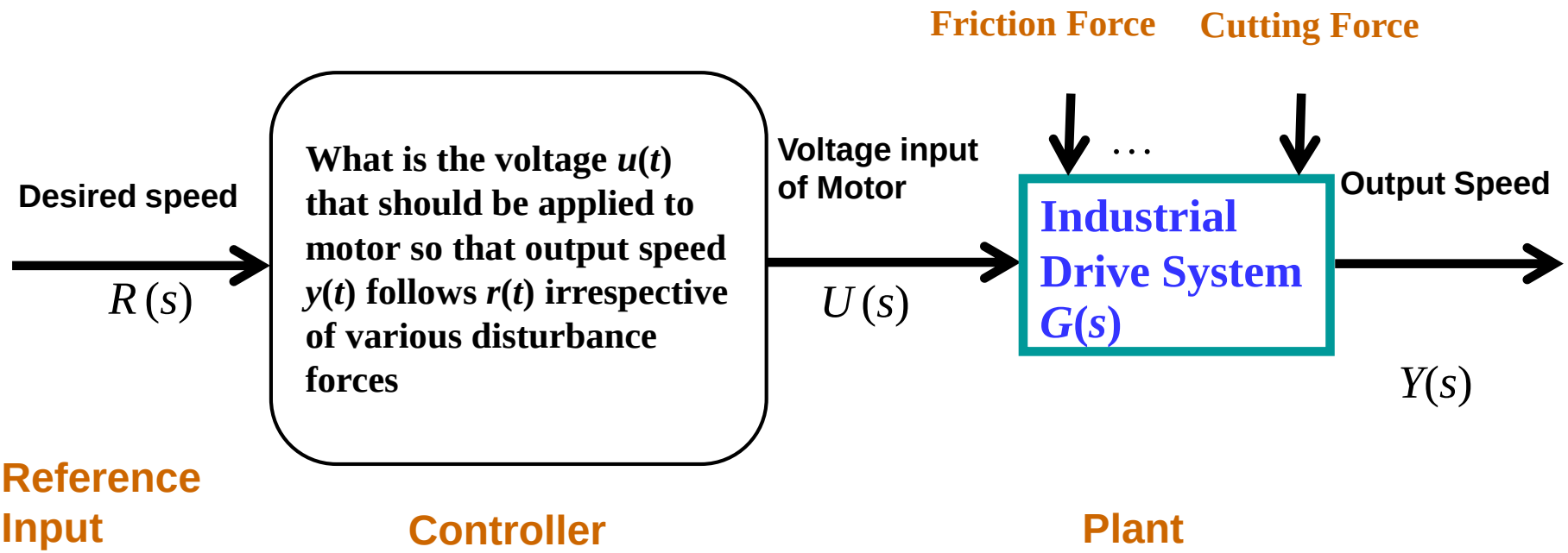
Introduction of Feedback Principle

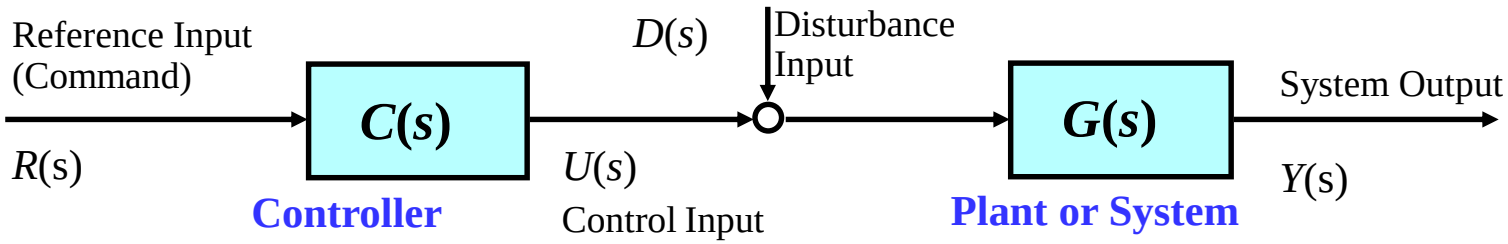
- **Objectives of Control System Design**
  - Open-loop control vs. Close-loop control
- **Objectives of Feedback and Feedforward Control**
- **General Control Design Principles**
  - Process inverse: High-gain Control, Model Compensation by Feedforward
- **General Structure of Controller**

Object of Control System Design



Open-Loop vs Closed-Loop

➢ Open-Loop Control



Q:Ideally, if we want Y(s) to follow R(s) (i.e. want Y(s) = R(s)), how would you design the controller C(s) for the above open-loop control system?（设计开环控制器使输出*Y*(*s*)跟踪*R*(*s*)）

Want

$$Y(s) = G(s)[u(s) + D(s)]$$
$$= G(s)C(s)R(s) + G(s)D(s) = R(s)$$

$$\Rightarrow \quad C(s) = G^{-1}(s)$$

System inversion!

Q:Can we attenuate the effect of disturbance D(s) on system output Y(s)?（可以抑制干扰对输出的影响吗?）

No!

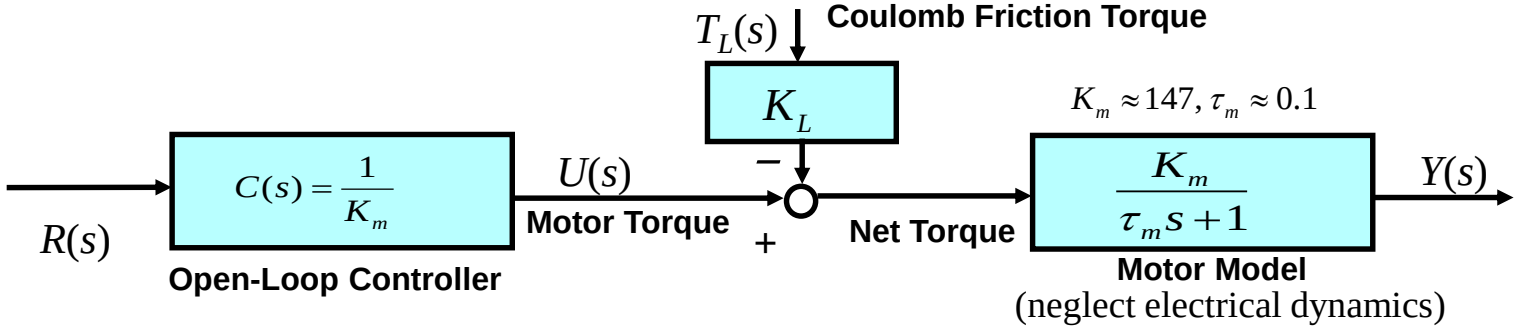
Q:Can we attenuate the effect of variations of plant transfer function G(s) on system output Y(s)?（可以抑制传递函数模型变化对输出的影响吗？）

No!

Open-Loop Control Example

➢ Speed Control of a DC Motor without Sensors

☐ Block Diagram of Open-Loop Control System



☐ Ideal Open-Loop Controller

$$C_{\text{ideal}}(s) = \frac{1}{G(s)} = \frac{\tau_m s + 1}{K_m} \Rightarrow U(s) = C_{\text{ideal}}(s)R(s) = \frac{\tau_m}{K_m} sR(s) + \frac{1}{K_m} R(s)$$

In time-domain: $u(t) = \frac{\tau_m}{K_m} \dot{r}(t) + \frac{1}{K_m} r(t)$

☐ An Implementable Open-Loop Controller

$$C(s) = \frac{1}{K_m} \Rightarrow U(s) = \frac{1}{K_m} R(s) \Leftrightarrow u(t) = \frac{1}{K_m} r(t)$$

☐ Output Speed with Open-Loop Controller

$$Y(s) = \frac{\cancel{K_m}}{\cancel{K_m}(\tau_m s + 1)} R(s) + \left[-\frac{K_L K_m}{\tau_m s + 1} \right] T_L(s)$$

Open-Loop Control Example

- Speed Control of a DC Motor without Sensors
  - ➢**Steady-state output speed for constant desired speed reference inputs of *r* (no friction)** (定常期望速度参考输入下的稳态输出输出(无摩擦))

$$T_L(s)=0 \Rightarrow Y(s)=\frac{1}{(\tau_m s+1)}R(s)$$

For constant desired speed $R(s)=r/s$

$$y_{ss}=\lim_{s\to 0} sY(s)=\lim_{s\to 0}\frac{1}{\tau_m s+1}sR(s)=r$$

Time constant (no controller) $\boxed{\tau_m}$

Time constant (with controller) $\boxed{\tau_m}$

No Change!

  - ➢**Steady-state output speed for constant desired speed reference inputs of *r* (with a constant friction τ_L)** (定常期望速度参考输入下的稳态输出输出(定常摩擦力力矩))

$$y_{ss}=\lim_{s\to 0} sY(s)=\lim_{s\to 0}\left[\frac{1}{(\tau_m s+1)}sR(s)-\frac{K_m K_L}{\tau_m s+1}sT_L(s)\right]$$

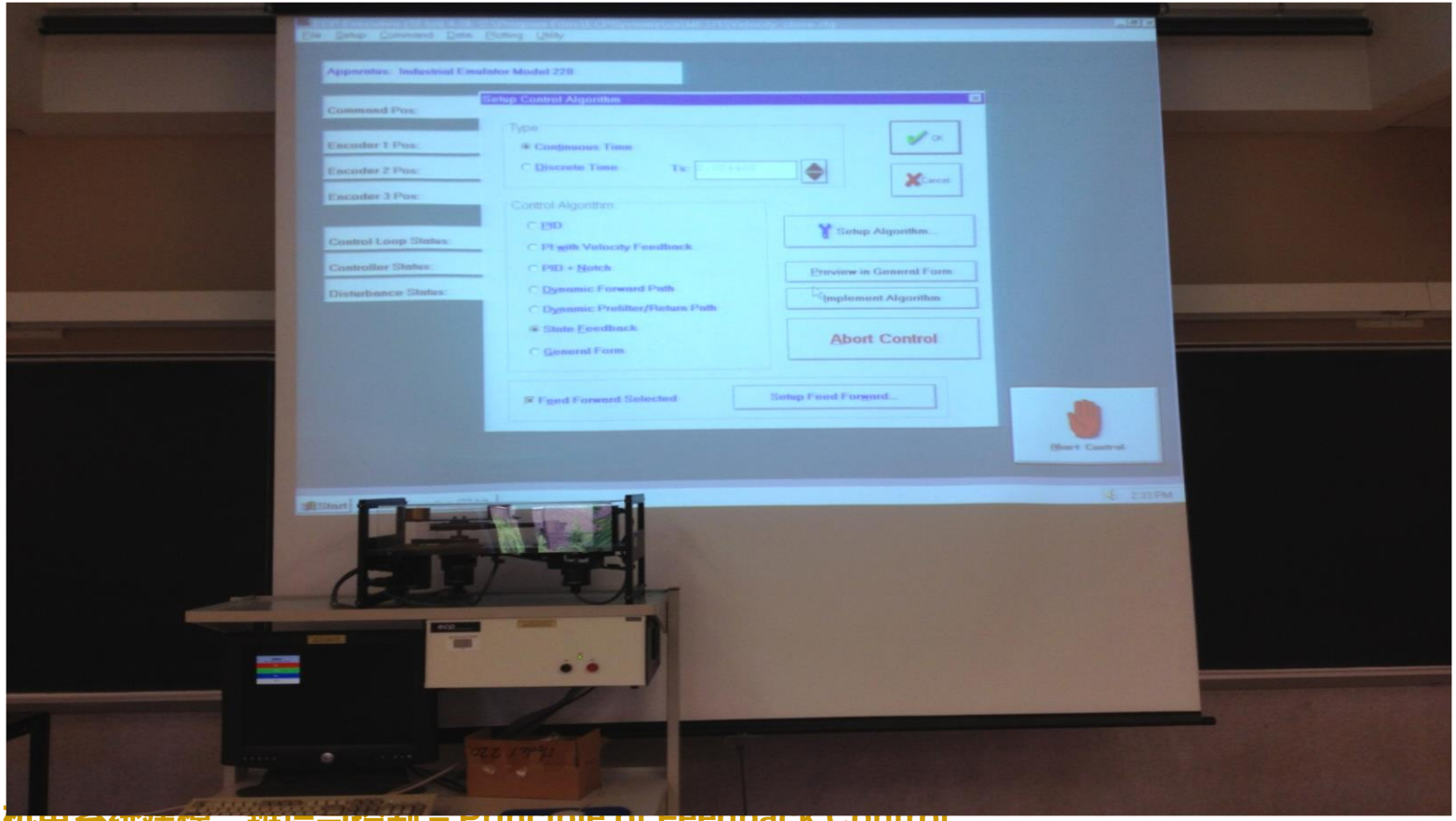
$$=r-K_m K_L \tau_L$$

There is no attenuation on the effect of friction !

Q: In reality, the friction on a motor may change quite significantly. Will the customer be happy with such an open-loop controller ? (实际作用在电机上的摩擦力可能变化较大，对这样的开环控制器满意吗?)

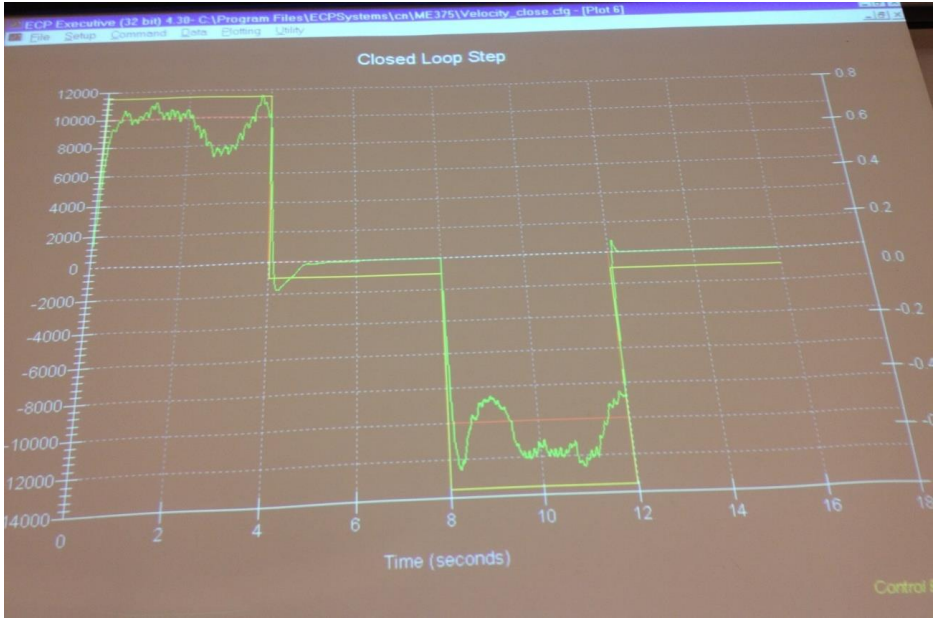
No!

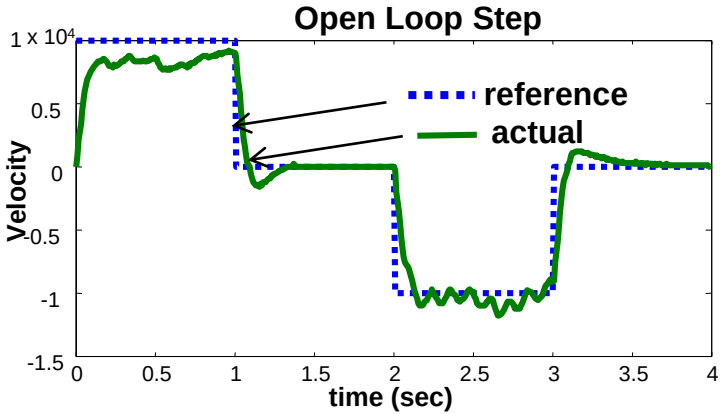
Experiments on Speed Control of a DC Motor

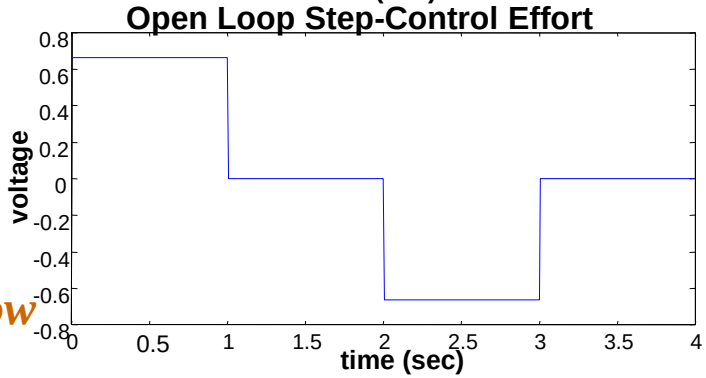


Open Loop Experimental Results

- **Speed Control of a DC Motor without Sensors**



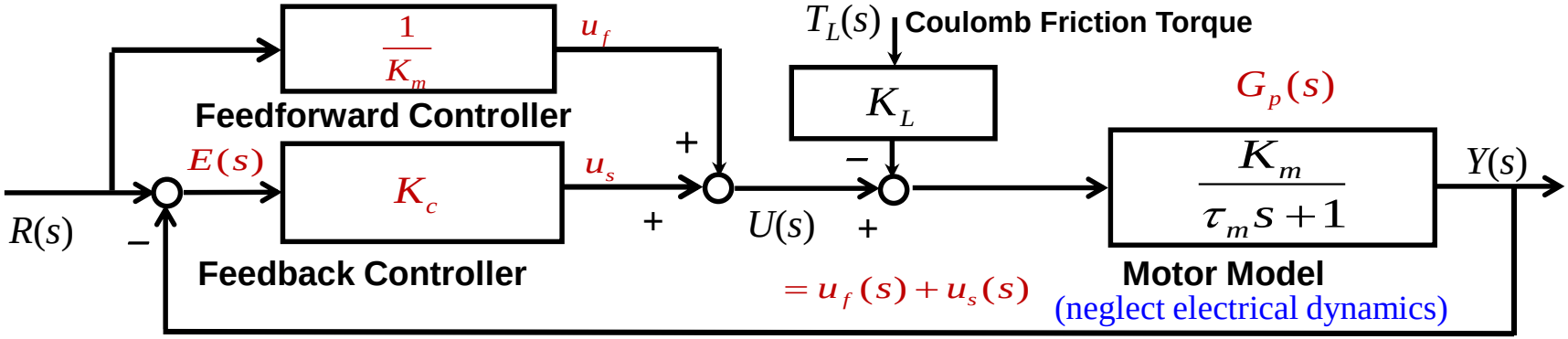




Q: Can you explain the experimental results obtained (e.g., the quality of the tracking performance and how it is different from the theoretical prediction)?

Closed-Loop Control Example

- **Speed Control of a DC Motor with Speed Sensors**



Block Diagram of A Simple Closed-Loop Control System

- **A Simple Closed-Loop Controller**
  - **Feedforward Controller**

$$u_f(s) = \frac{1}{K_m} R(s) \quad \Leftrightarrow \quad u_f(t) = \frac{1}{K_m} r(t)$$

  - **Feedback Controller**

$$u_s(s) = K_c E(s) = K_c[R(s) - Y(s)] \quad \Leftrightarrow \quad u_s(t) = K_c[r(t) - y(t)]$$

Closed-Loop Control Example

- **Speed Control of a DC Motor with Speed Sensors**
  - **Output speed with closed-loop controller**

$$Y(s)=\underbrace{\frac{1+K_mK_c}{\tau_m s+1+K_mK_c}}_{G_{YR}(s)}R(s)-\frac{K_LK_m}{\tau_m s+1+K_mK_c}T_L(s)$$

Closed-loop time constant:

$$\boxed{\tau=\frac{\tau_m}{1+K_mK_c}}$$

$$Y(s)=\underbrace{\frac{1}{\frac{\tau_m}{1+K_mK_c}s+1}}_{G_{YR}(s)}R(s)+\underbrace{\left[-\frac{\frac{K_LK_m}{1+K_mK_c}}{\frac{\tau_m}{1+K_mK_c}s+1}\right]}_{G_{YT}(s)}T_L(s)$$

  - **Steady-state (SS) output speed for constant desired speed reference inputs of *r* (no friction)**

$$T_L(s)=0 \Longrightarrow y_{ss}=G_{YR}(0)r=r$$

No SS speed error !

  - **Steady-state output speed for constant desired speed reference inputs *r* (with a constant friction τ_L)**

$$y_{ss}=G_{YR}(0)r+G_{YT}(0)\tau_L=r-\frac{K_LK_m}{1+K_mK_c}\tau_L$$

Closed-Loop Control Example

- **Speed Control of a DC Motor with Speed Sensors**
  - **Steady-state output speed error for constant desired speed reference inputs of *r* (with a constant friction τ_L)**

$$e_{ss} = r - y_{ss} = \frac{K_L K_m}{1 + K_m K_c}\tau_L$$

decrease with increasing K_c!

Q: When can we consistently have the desired steady-state speed regardless certain amount of Coulomb friction that may exist ?（什么时候能持续获得期望稳态速度而不管摩擦力有多少）

Using large enough feedback gain K_c so that the effect of disturbances like the Coulomb friction is attenuated sufficiently（采用足够大的反馈增益以便使像摩擦力那样的干扰能被足够抑制）

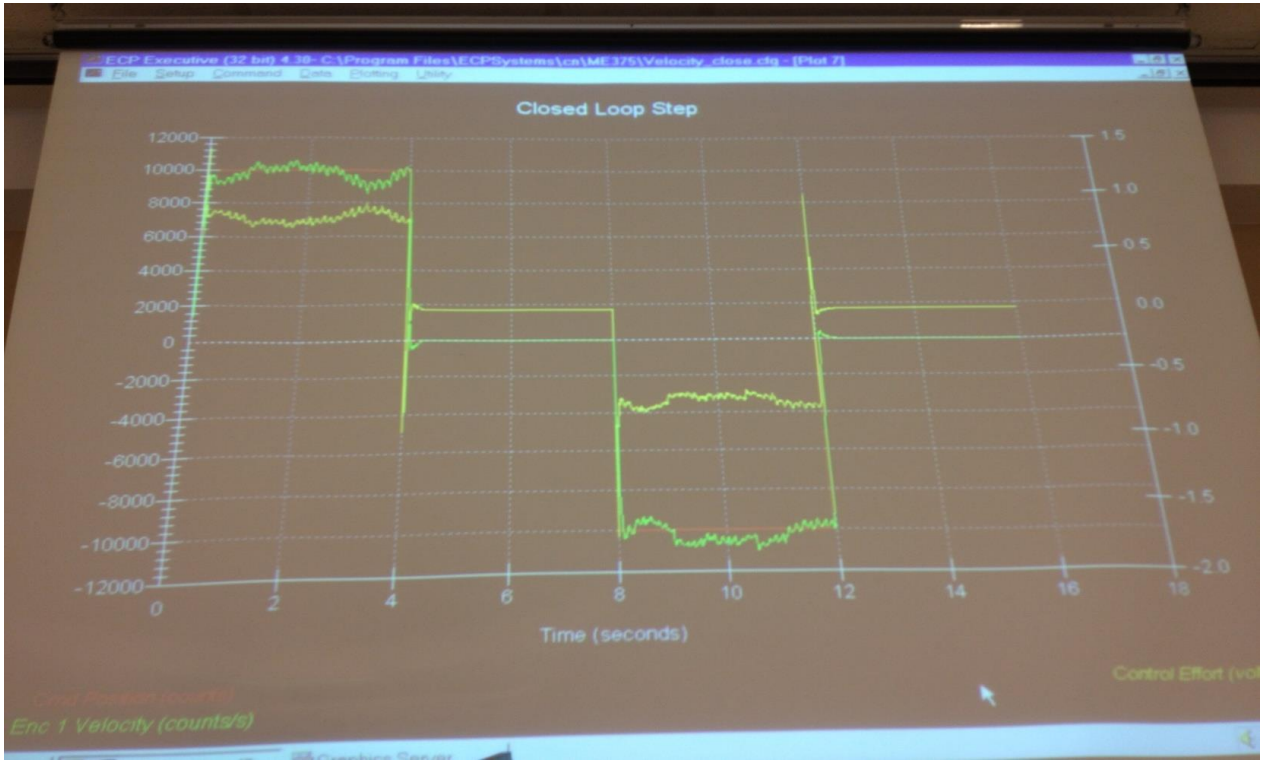
Q: How is response speed of the closed-loop system compared with the response speed of original open-loop system ?（相比开环系统，闭环系统的响应速度如何）

Faster as the CL time constant $\tau = \frac{\tau_m}{1 + K_m K_c}$ is smaller than open-loop time constant τ_m

（快！由于闭环系统时间常数比开环小）

Closed-Loop Experimental Results

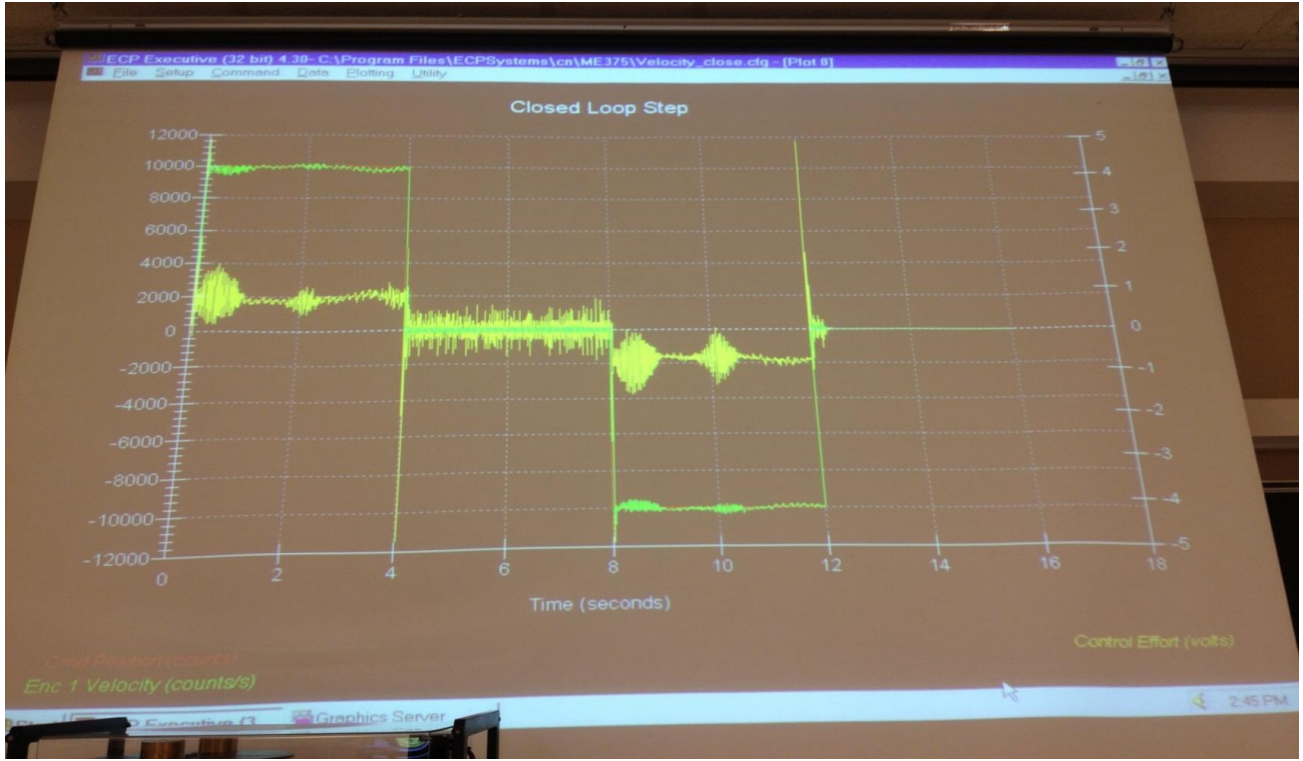
- **Speed Control of a DC Motor with Sensors and Feedforward**



K_c=0.01

Closed-Loop Experimental Results

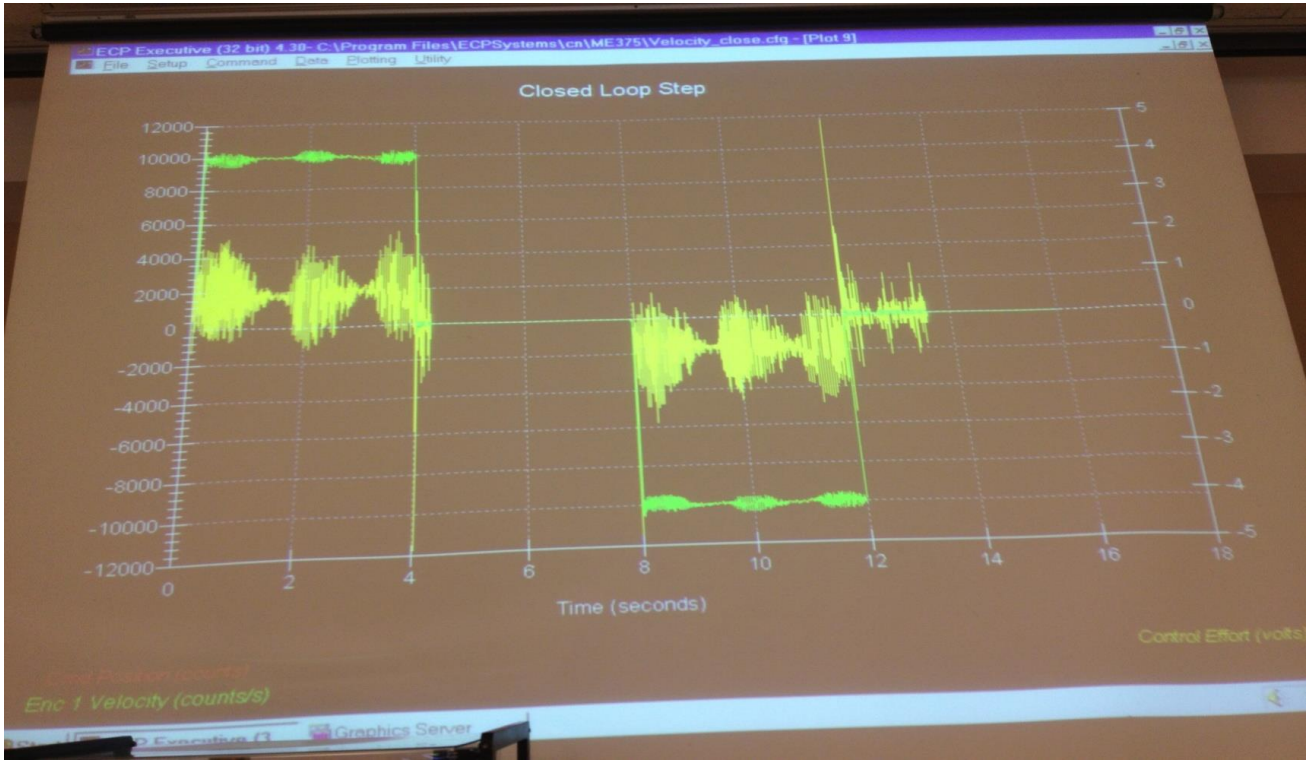
- **Speed Control of a DC Motor with Sensors and Feedforward**



K_c=0.07

Closed-Loop Experimental Results

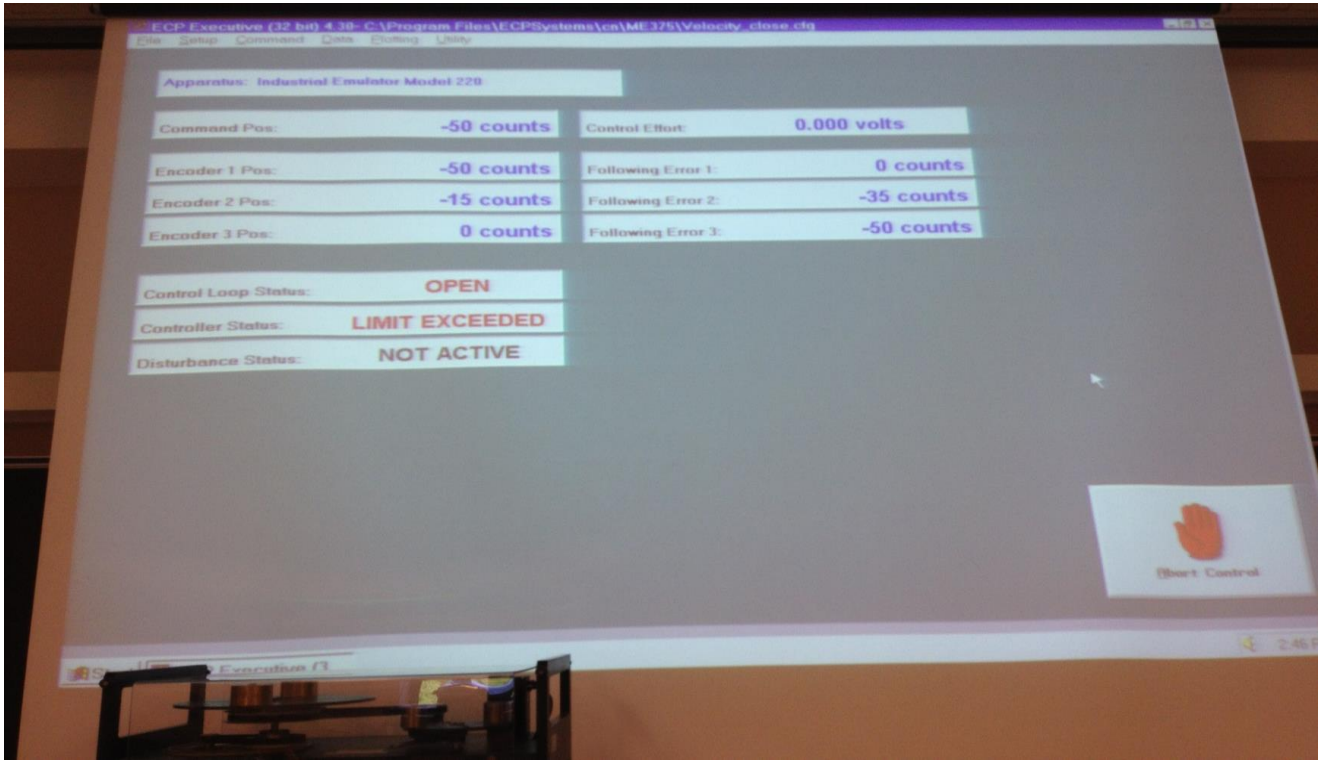
- **Speed Control of a DC Motor with Sensors and Feedforward**



K_c=0.11

Closed-Loop Experimental Results

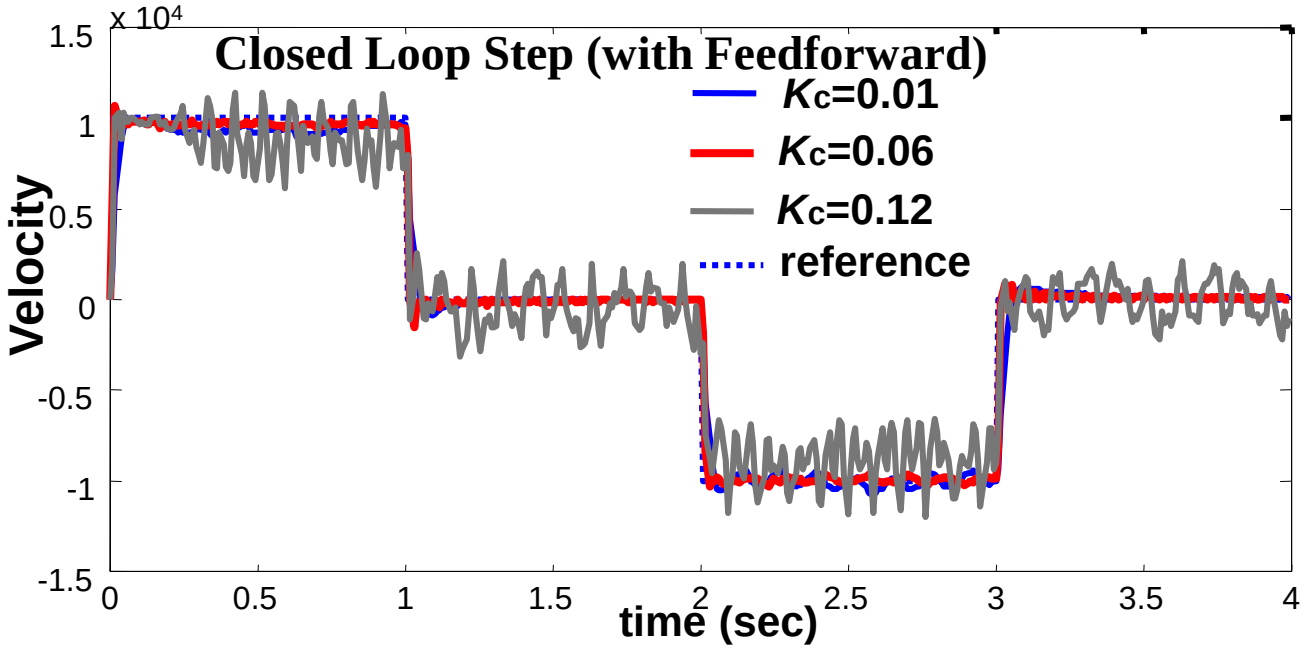
- **Speed Control of a DC Motor with Sensors and Feedforward**



K_c=0.13

Closed-Loop Experimental Results

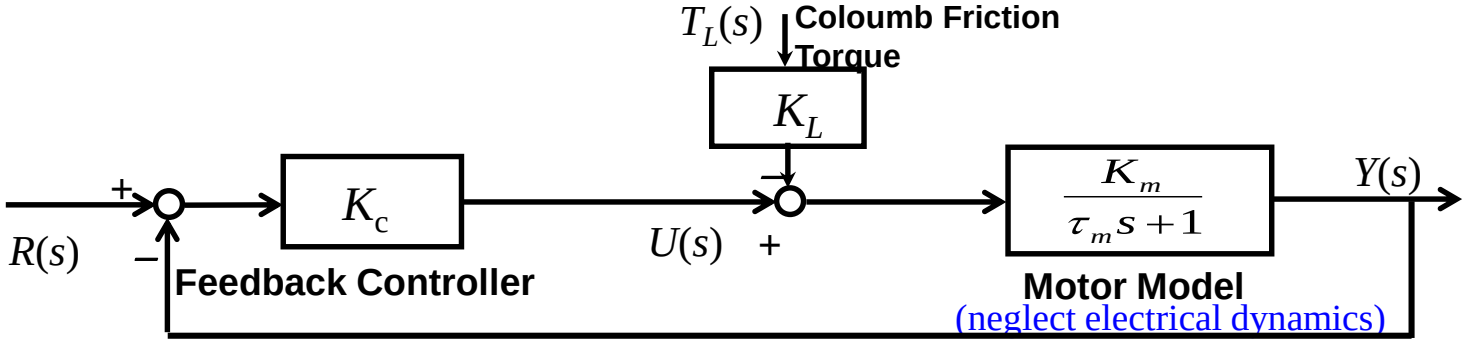
➢ **Speed Control of a DC Motor with Sensors and Feedforward**



Q: *Can you explain the experimental results obtained (e.g., how the quality of the tracking performance and response speed change with feedback gain as predicted by theory)?*

Closed-Loop Control Example

- **Speed Control of a DC Motor with Feedback Action Only**



Closed-loop Controller

Block Diagram of A Simple Closed-Loop Control System (No feedforward)

$$Y(s)=\underbrace{\frac{K_m}{\tau_m s+1}}_{G_p(s)}\left[U(s)-K_L T_L(s)\right]=G_p(s)\left[K_c R(s)-K_c Y(s)-K_L T_L(s)\right]$$

$$\Rightarrow \quad Y(s)=\frac{G_p(s)K_c}{1+K_c G_p(s)}R(s)+\left[\frac{-K_L G_p(s)}{1+K_c G_p(s)}T_L(s)\right]=\frac{\frac{K_m}{\tau_m s+1}K_c}{1+K_c\frac{K_m}{\tau_m s+1}}R(s)-\frac{K_L\frac{K_m}{\tau_m s+1}}{1+K_c\frac{K_m}{\tau_m s+1}}T_L(s)$$

$$E(s)=R(s)-Y(s)=\frac{\frac{1}{1+K_m K_c}}{\frac{\tau_m}{1+K_m K_c}s+1}R(s)+\frac{\frac{K_m K_L}{1+K_m K_c}}{\frac{\tau_m}{1+K_m K_c}s+1}T_L(s)$$

$$y_{ss}=G_{YR}(0)r+G_{YT}(0)\tau_L=\frac{K_m K_c}{1+K_m K_c}r-\frac{K_L K_m}{1+K_m K_c}\tau_L$$

$$e_{ss}=r-y_{ss}=\frac{1}{1+K_m K_c}r+\frac{K_L K_m}{1+K_m K_c}\tau_L$$

Feedforward vs Feedback Control

- **Speed Control of a DC Motor with Sensors**

☐ **With Feedback and Feedforward :**

$$Y(s)=\frac{1}{\frac{\tau_m}{1+K_mK_c}s+1}R(s)-\frac{\frac{K_LK_m}{1+K_mK_c}}{\frac{\tau_m}{1+K_mK_c}s+1}T_L(s)$$

$$e_{ss}=r-y_{ss}=\frac{K_LK_m}{1+K_mK_c}\tau_L$$

☐ **With Feedback Action Only :**

$$Y(s)=\frac{\frac{K_mK_c}{1+K_mK_c}}{\frac{\tau_m}{1+K_cK_m}s+1}R(s)-\frac{\frac{K_mK_L}{1+K_cK_m}}{\frac{\tau_m}{1+K_cK_m}s+1}T_L(s)$$

$$e_{ss}=r-y_{ss}=\frac{1}{1+K_mK_c}r+\frac{K_LK_m}{1+K_mK_c}\tau_L$$

Q: How is the performance of the closed-loop control with feedback control action only compared to that of the closed-loop control with both feedforward and feedback actions ?

☐ **Attenuation ability of the system to disturbances such as the Coulomb friction :**

Same

☐ **Response speed:**

$$\tau_m'=\frac{\tau_m}{1+K_mK_c}$$

Same

☐ **Steady-state output speed error for unit constant desired speed without considering friction effect :**

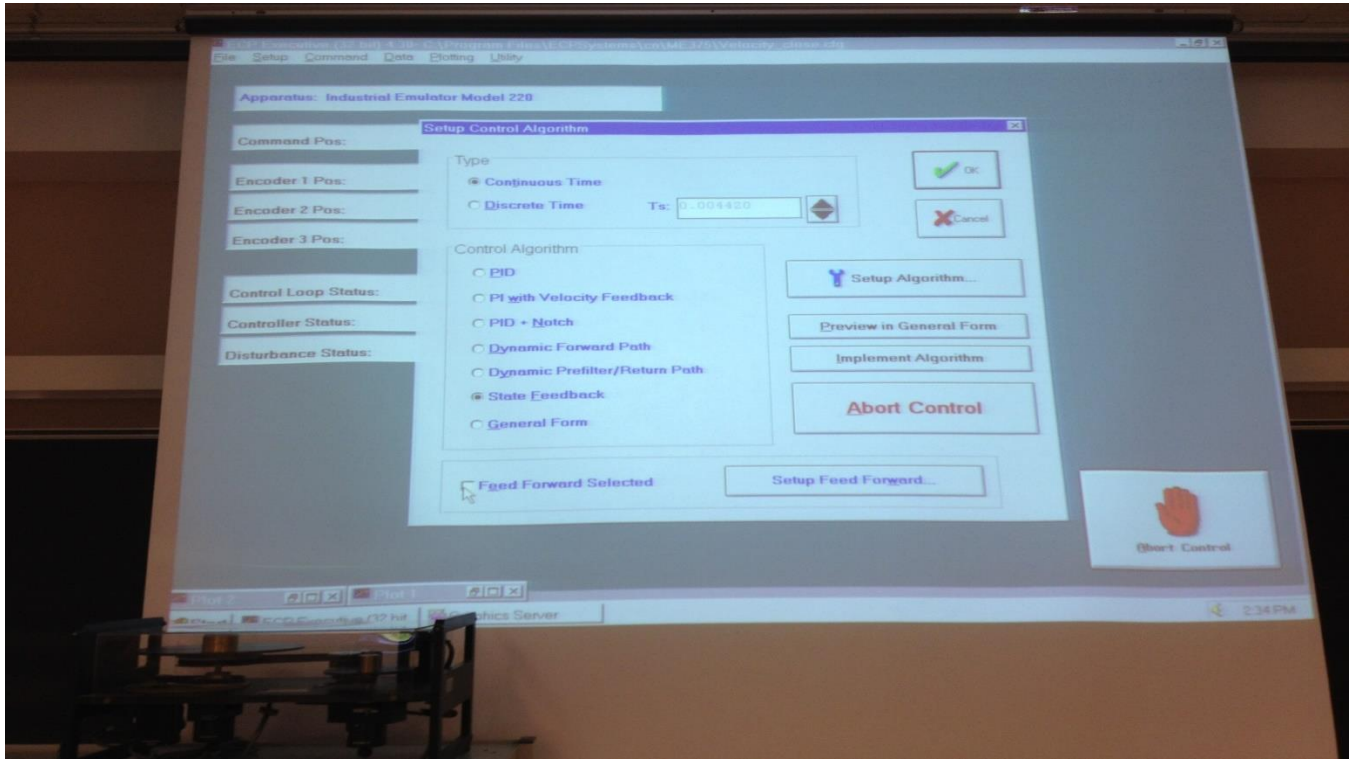
A little bit larger for Feedback Only

when K_c is very large

$$\frac{1}{1+K_mK_c}\approx 0$$

Closed-Loop Experimental Results

- **Speed Control of a DC Motor with Sensors but no Feedforward**



Closed-Loop Experimental Results

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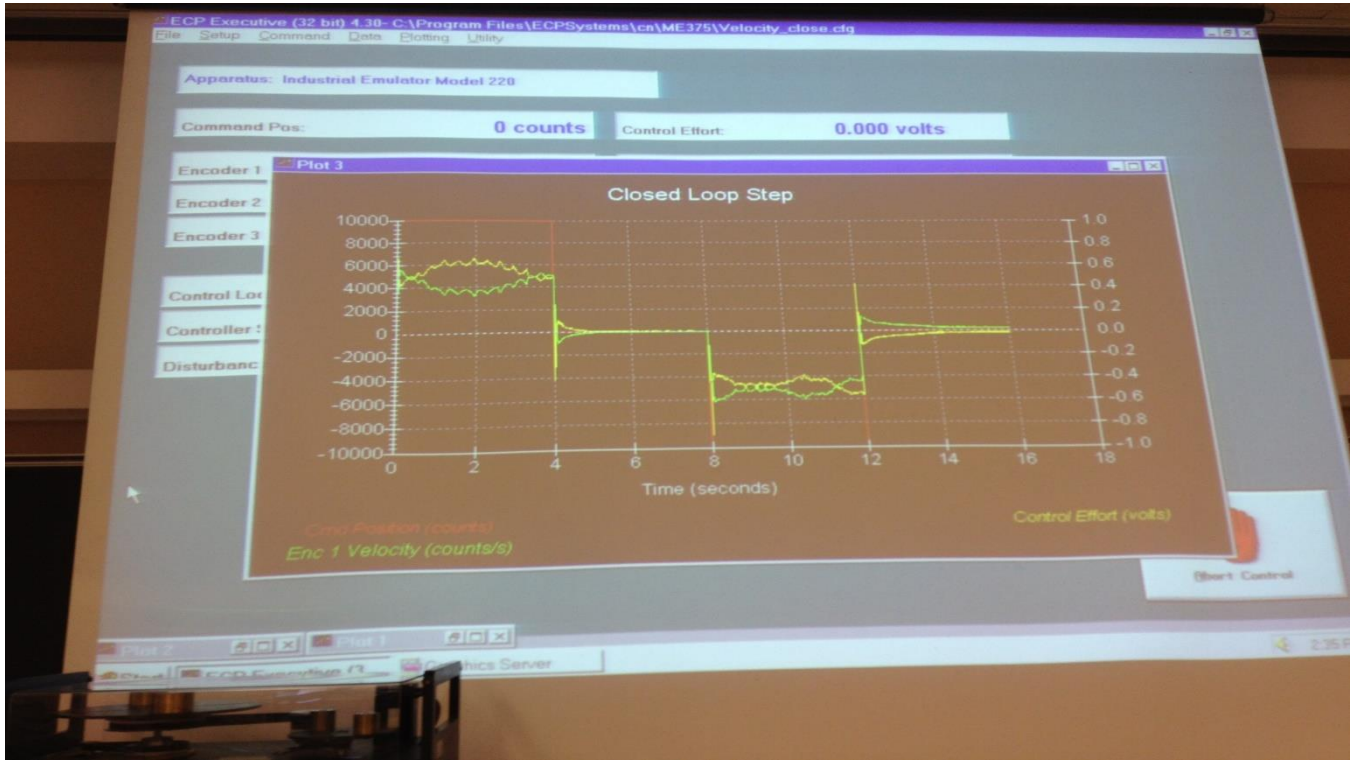
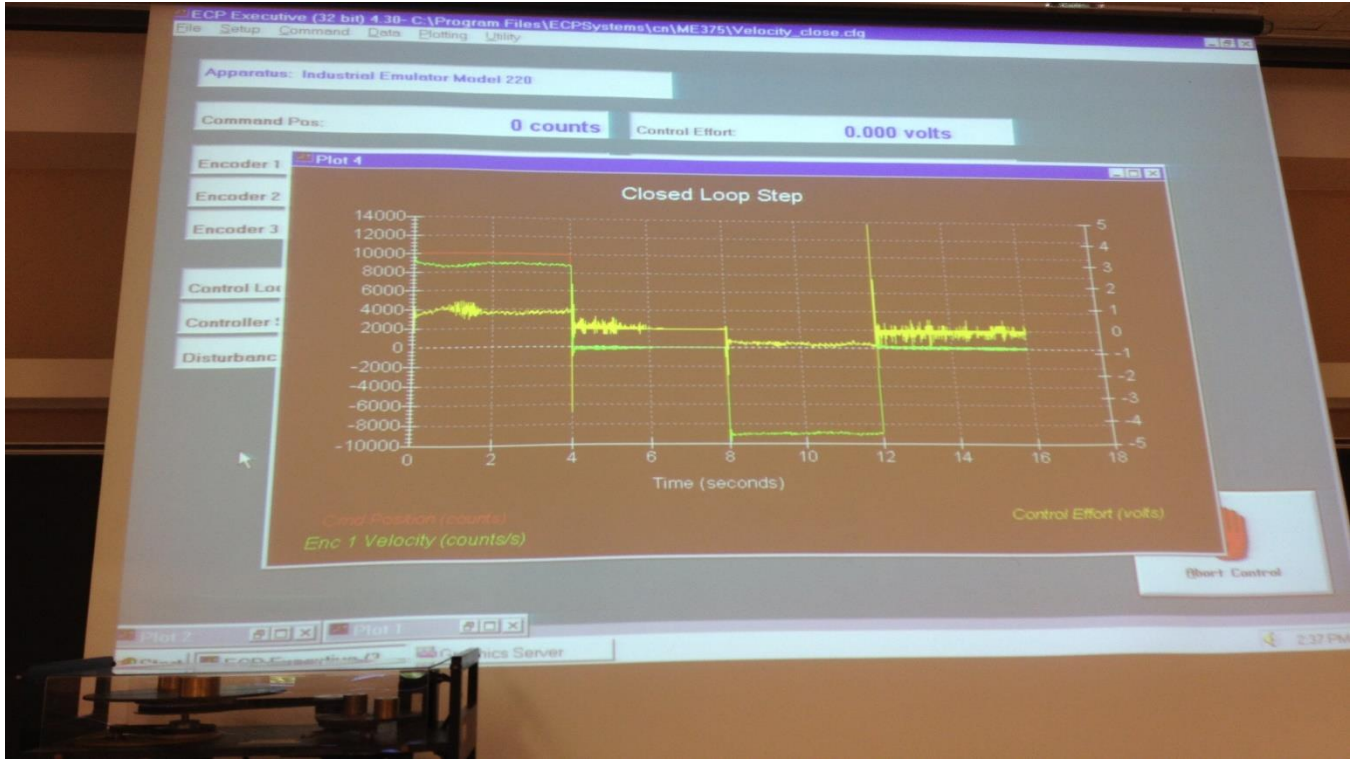


K_c=0.01

Closed-Loop Experimental Results

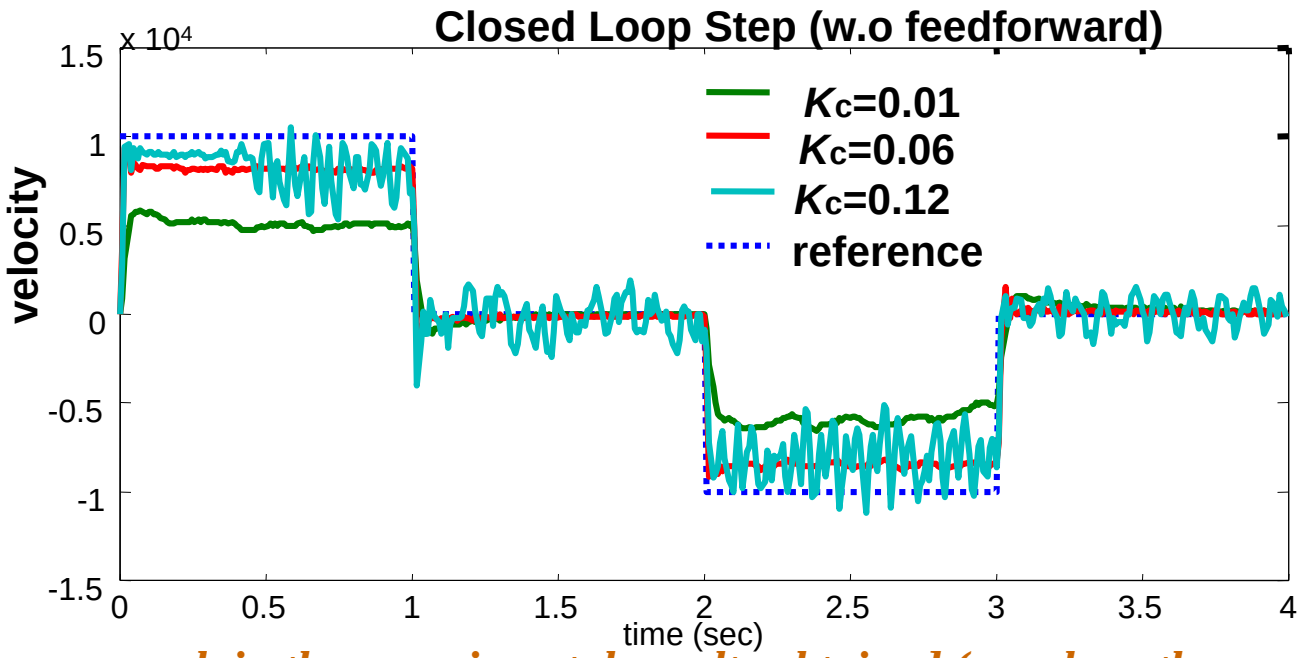
- Speed Control of a DC Motor with Sensors but no Feedforward



K_c=0.07

Closed-Loop Experimental Results

- **Speed Control of a DC Motor with Feedback Action Only**



Q: Can you explain the experimental results obtained (e.g., how the quality of the tracking performance and response speed change with feedback gain as predicted by theory)?

Why Feedback ?

➢ Objectives of Feedback

- **Stabilize unstable systems** (稳定不稳定的系统)
  - ✓ Adjust the closed-loop system's dynamic behavior. For example, unstable plants such as inverted pendulum and the position control of DC motor (调节闭环系统的动态行为，如倒立摆控制或直流电机位置控制)
- **Improve system performance to meet stringent performance requirements** (提高具有严格要求的系统性能)
  - ✓ Steady State Performance: for example, reduce steady state error due to disturbances ...(稳态性能，如降低干扰引起的稳态误差)
  - ✓ Transient Performance: for example, reduce rise time and settling time to speed up system response , …(瞬态性能，如降低上升时间和调整时间以加速系统响应)
- **Reduce the effect of modeling uncertainty and various disturbances** (降低模型干扰和不确定)
  - ✓ Employing high gain feedback at the right location and frequency range (在适当的位置和频段采用高增益反馈)
  - ✓ Possible instability of closed-loop system in the presence of various physical constraints (在各种无力限制存在时闭环系统可能的不稳定性)

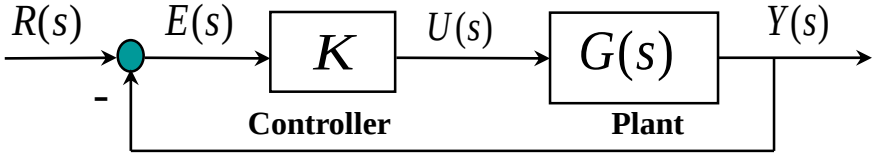
Why Feedforwad ?

- **Model Compensation in Advanced Nonlinear Controls** (在先进非线性控制中的模型补偿)
  - Makes the control input close to the desired control input that is needed to accomplish the task (使控制输入靠近完成任务所需的期望控制输入)
  - Only small amount of control correction needs to be provided by feedback control action, which results in smaller tracking error (keep in mind that feedback control needs tracking error to generate the control action) (仅用反馈控制提供小幅控制修正以获得较小跟踪误差(注意反馈控制需要跟踪误差来产生控制行为))

- **Applications of Feedforward**
  - For applications having stringent performance requirements such as control of precision electro-mechanical devices (具有严格性能要求的应用，如精密机电设备)
  - Usefulness of feedforward heavily depends on the accuracy of models used for physical plants, which normally have quite large variations of system parameters (前馈的有效性严重依赖于物理系统模型精度,其参数可能有较大变化)
  -)

General Control Design Principles

- **Control design is nothing but an *Inversion Process*** (控制设计没什么就是求逆)
- **The inversion can be achieved by two key mechanisms** (求逆有两种方法)
  - **(High-Gain) Feedback and**



$$\frac{U(s)}{R(s)} = \frac{K}{1+KG} \overset{\text{large } K}{\approx} G^{-1}$$

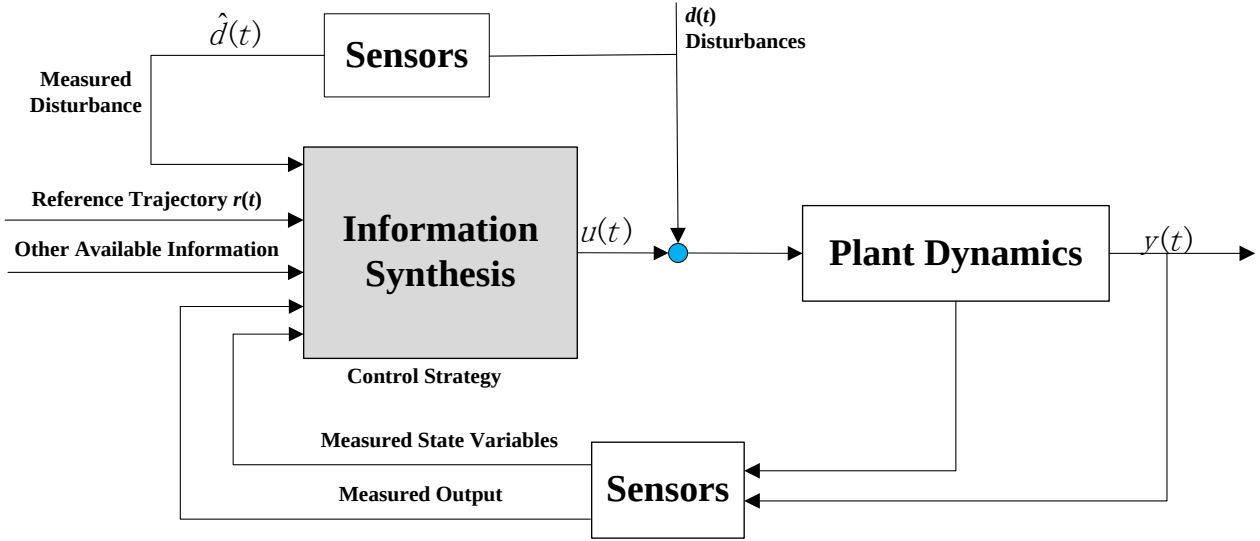
high-gain (高增益) → **high sensitivity to measurement noises** (对噪声高度敏感) → **makes the stability of closed-loop system sensitive to control input saturation and neglected high-frequency dynamics** (使闭环系统的稳定性对控制输入饱和和忽略的高频动力学较为敏感)

  - **(Model Compensation) Feedforward**

On-line learning (在线学习) → **A key to have a good model compensation of feedforward design** (前馈设计获得较好模型补偿的关键)

General Structure of Controller

In general, a controller is nothing but a strategy to determine a control action based on all available information; information not only comes from the measured output but also from the measured internal state variables, measured disturbance, reference trajectory, and plant model structure（通常，控制器就是基于所有可得信息来决定一个控制行为的策略，这些信息不仅来自于测量输出，也来自于测量内部状态，测量干扰，参考轨迹和模型结构）



So use your own imagination to come out new control schemes!